



The Deepest GLIMPSE of a Dense Gas Cocoon Enshrouding a Little Red Dot

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Abstract

The detection of strong Balmer breaks and absorption features in Little Red Dots (LRDs) suggests they host active galactic nuclei embedded within dense gas envelopes, potentially powered by super-Eddington accretion. We present GLIMPSE-17775, a luminous ($L_{\text{bol}} \sim 10^{45} \text{ erg s}^{-1}$) LRD at $z = 3.501$ behind Abell S1063 ($\mu \sim 2$), observed with deep JWST/NIRCam and a ~ 20 hr (80 hr delensed) NIRSpect G395M spectrum. The data reveal over 40 emission and absorption features, including a rich forest of low-ionization Fe II lines and numerous broad hydrogen recombination transitions. We use this depth to test the dense-gas interpretation through five independent diagnostics. Nearly all permitted lines show exponential wings with consistent FWHM, a signature of Thomson scattering requiring $n_e \gtrsim 10^8 \text{ cm}^{-3}$. Adopting this width yields $M_{\text{BH}} \sim 10^{6.7} M_{\odot}$, a factor of 10 lower than Gaussian fits, and $\lambda_{\text{Edd}} \sim 1.8$. Additional diagnostics support the same picture: a pronounced Balmer break ($f_{\nu,4050}/f_{\nu,3670} = 2.0 \pm 0.1$), enhanced He I $\lambda 7065$ and $\lambda 10830$ with P-Cygni absorption, Bowen-fluorescent O I $\lambda 8446$ – $\lambda 11290$ emission requiring Ly β pumping, and 16 Fe II lines matching fluorescence models. These features indicate a dense ($n \sim 10^8 \text{ cm}^{-3}$), partially ionized cocoon where scattering and fluorescence dominate line formation, providing strong evidence that at least some LRDs are powered by super-Eddington black hole growth in the early Universe.

Unified Astronomy Thesaurus concepts: Black holes (162); Active galactic nuclei (16); Galaxy evolution (594)

1. Introduction

One of the most enticing puzzles brought about by the launch of the James Webb Space Telescope (JWST) has been the discovery of red, compact objects called “Little Red Dots” (LRDs; J. Matthee et al. 2024). Previously invisible to the Hubble Space Telescope (HST) due to their extreme faintness in the optical and lack of near-infrared (NIR) coverage, LRDs have emerged in abundance (V. Kokorev et al. 2024a; H. B. Akins et al. 2025a; D. D. Kocevski et al. 2025; G. Barro et al. 2026) thanks to the unprecedented NIR sensitivity of JWST.

Their unusual properties—such as compact morphologies in rest-optical and distinctive “V-shaped” spectral energy distributions (SEDs)—make LRDs easy to identify in JWST fields; however, this is where the simplicity ends. Explaining

LRDs as either evolved or dusty compact galaxies proves difficult: The former scenario requires massive stellar populations that strain Lambda cold dark matter (Λ CDM) predictions (M. Boylan-Kolchin 2023; I. Labbé et al. 2023), while the latter implies significant dust emission, yet none is observed (C. M. Casey et al. 2024; G. C. K. Leung et al. 2025; H. B. Akins et al. 2025a; I. Labbé et al. 2025).

Over time, the accumulating detections of broad Balmer-series emission lines (D. D. Kocevski et al. 2023, 2025; J. Matthee et al. 2024, just to name a few), often accompanied by signatures of extreme ionization (e.g., V. Kokorev et al. 2023), have begun to clarify the physical origin of LRDs, pointing increasingly toward active galactic nuclei (AGN) as the underlying power source. In parallel, the much needed advent of NIRSpect Micro Shutter Assembly (MSA) programs based on red targets selected from JWST imaging (e.g., A. de Graaff et al. 2025b) has provided critical confirmation: Nearly all point sources exhibiting “V-shaped” SEDs reveal broad emission lines upon spectroscopic follow-up (R. E. Hviding et al. 2025), solidifying their AGN interpretation.



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The nature of the spectral inflection point in LRDs, typically located near ~ 3600 Å, has also undergone significant revision. Initially interpreted as a stellar Balmer break (I. Labbé et al. 2023), this feature implied implausibly high stellar masses (M_*) far too early in cosmic history (M. Boylan-Kolchin 2023; N. Sabti et al. 2024). A second hypothesis invoked differential dust attenuation and host-galaxy contamination in the rest-UV to explain the sharp discontinuity (M. Volonteri et al. 2025). However, this explanation was soon ruled out by D. J. Setton et al. (2025) and Y. Ma et al. (2025), who demonstrated that the break generally lies around the Balmer limit, albeit with a fundamentally different origin than initially proposed.

More recently, a series of theoretical and observational works have converged on a new picture in which LRDs may host accreting black holes enshrouded in exceptionally dense, partially ionized gas cocoons, often referred to as “black hole stars” (hereafter BH^{*}; A. de Graaff et al. 2025b; K. Inayoshi & R. Maiolino 2025; R. P. Naidu et al. 2025; A. J. Taylor et al. 2025). Here, BH^{*} is used in an empirical sense to denote cocooned black holes whose emergent spectra exhibit a combination of black-hole-like and star-like features, such as broad permitted lines, steep Balmer-limit breaks, and Balmer absorption, as observed in the archetypal BH^{*} sources (A. de Graaff et al. 2025b; R. P. Naidu et al. 2025). This usage does not assume a specific geometry or formation channel: The underlying physical picture involves an AGN power-law ionizing spectrum processed through very dense gas, and similar phenomenology can arise in a range of dense-gas configurations (e.g., K. Inayoshi & R. Maiolino 2025; D. Kido et al. 2025; H. Liu et al. 2025; A. Sneppen et al. 2026). In such environments, suppressed X-ray and radio emission (e.g., K. Inayoshi & R. Maiolino 2025; R. P. Naidu et al. 2025) arise naturally from the high optical depth, while the UV emission and forbidden optical lines (e.g., [O III] $\lambda\lambda 4959, 5007$) may originate in the more extended host (e.g., A. de Graaff et al. 2025d).

The dense-gas interpretation also offers a natural explanation for the P-Cygni-like Balmer and helium profiles frequently seen in high-signal-to-noise-ratio (S/N) spectra (J. Matthee et al. 2024), the non-Gaussian, exponentially winged line shapes predicted by radiative-transfer models of scattering in ionized gas (S.-J. Chang et al. 2026; V. Rusakov et al. 2026), and Ly α -like resonantly scattered shapes (R. P. Naidu et al. 2025; S.-J. Chang et al. 2026). Intriguingly, if electron and resonant scattering indeed dominate the broad-line widths, then the true virial velocities (traced by intrinsically narrower Gaussian cores)—and hence black hole masses—could be lower by an order of magnitude, alleviating the apparent tension between LRD black hole masses and their compact host galaxies (V. Rusakov et al. 2026).

So far, however, the BH^{*}/dense-gas interpretation of LRDs has relied largely on the Balmer break, occasional absorption features (X. Lin et al. 2026), and, to a limited extent, the shapes of broad emission lines (V. Rusakov et al. 2026). While these features are consistent with the presence of dense, partially ionized gas, they stop short of providing direct spectroscopic confirmation of the physical conditions expected by the BH^{*} scenario. What has been missing is an unambiguous demonstration—through emission-line physics—that LRDs indeed host dense, optically thick cocoons surrounding rapidly accreting black holes. Such evidence would directly tie the observed line formation, excitation, and

radiative transfer to dense, stratified envelopes of gas surrounding the central engine.

In this work, we present precisely such a case: an exceptionally deep, ~ 20 hr (equivalent to 80 hr without lensing magnification) JWST/NIRSpec G395M spectrum of a luminous LRD at $z = 3.50102$. The target, GLIMPSE-17775, lies in a highly magnified region of the massive galaxy cluster Abell S1063 (hereafter AS1063) and benefits from the combined power of JWST/NIRCam imaging from the GLIMPSE GO program (PID 3293; PIs: H. Atek & J. Chisholm) and recent NIRSpec spectroscopy from the GLIMPSEDirector’s Discretionary Time (DDT) (hereafter GLIMPSE-D) program (PID 9223; PIs: S. Fujimoto & R. Naidu). This synergy of ultra-deep spectroscopy and strong gravitational lensing enables an unprecedented view of GLIMPSE-17775, revealing rest-frame optical and NIR features at a level unseen before in any LRD. We detect over 40 emission and absorption features, including 16 Fe II transitions that form a dense “iron forest.” The remarkable richness of this spectrum makes GLIMPSE-17775 a uniquely powerful laboratory for dissecting the dense, radiation-dominated environments that accompany early black hole growth.

This paper is organized as follows. In Section 2, we present the GLIMPSE-D NIRSpec dataset alongside the photometric datasets used in this study. In Section 3, we calculate the spectroscopic redshift, describe the identification of prominent emission/absorption features, and describe bespoke fitting of various line complexes. Section 4 describes the measurement of the source morphology, dust attenuation, and black hole masses. In Section 5, we comment on the various properties of the emission features in the our target. Finally, we discuss our findings in Section 6.

Throughout this work, we assume a flat Λ CDM cosmology with $\Omega_{m,0} = 0.3$, $\Omega_{\Lambda,0} = 0.7$, and $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, and a G. Chabrier (2003) initial mass function between 0.1 and $100 M_{\odot}$. All magnitudes are expressed in the AB system (J. B. Oke 1974).

2. Observations and Data

The target, GLIMPSE-17775, was originally identified as a bright LRD candidate in JWST/NIRCam imaging captured by the GLIMPSE (PID: 3293; PIs: H. Atek & J. Chisholm) survey (H. Atek et al. 2025) of the lensed AS1063 Hubble Frontier Field (HFF; J. M. Lotz et al. 2017). The highly magnified area of AS1063 has already successfully yielded $z > 16$ galaxy candidates (V. Kokorev et al. 2025), potential Population III (Pop III) hosts (S. Fujimoto et al. 2025a), numerous faint and high-redshift galaxies (I. Chemerynska et al. 2026), new constraints on reionization (D. Korber et al. 2025), identification of intermediate-mass black holes (Q. Fei et al. 2025), and a wide variety of enigmatic LRDs, some moderately lensed. The LRD selection was based on the standard compactness plus “V-shape” criteria already laid out in J. E. Greene et al. (2024), V. Kokorev et al. (2024a), H. B. Akims et al. (2025a), and I. Labbé et al. (2025) using photometric redshifts derived with EAZY (G. B. Brammer et al. 2008), a technique that has proven successful at consistently identifying many exciting sources (see, e.g., H. B. Akims et al. 2025b; A. de Graaff et al. 2025b; V. Kokorev et al. 2023; R. P. Naidu et al. 2025; A. J. Taylor et al. 2025). Located at $z_{\text{phot}} = 4.2 \pm 0.1$, GLIMPSE-17775 stood out in particular due to its extreme rest-optical brightness with $m_{f444w} \sim 23.6$ mag and a modest

lensing magnification $\mu \sim 2$. While very little could be done with photometry alone, this marked GLIMPSE-17775 as a promising target for any future spectroscopic follow-up.

2.1. Photometry

We use photometry both to select targets for the GLIMPSE-D NIRSpec observations and, during spectroscopic modeling, to constrain the overall shape of the SED and correct for NIRSpec slit losses. In addition to JWST/NIRCam data, we incorporate deep HST Advanced Camera for Surveys and Wide Field Camera 3 imaging from the HFF (J. M. Lotz et al. 2017) and BUFFALO (C. L. Steinhardt et al. 2020) programs. Our reprocessed HST mosaics are based on Gaia-aligned images from the CHArGE archive (V. Kokorev et al. 2022), hosted on the Dawn JWST Archive (F. Valentino et al. 2023). More descriptions of the image reduction and source extraction procedure can be found in R. Endsley et al. (2025), as well as in the GLIMPSE overview paper (H. Atek et al. 2025); we briefly summarize the latter procedure below.

Photometry is performed on point-spread function (PSF)-homogenized HST and JWST images, convolved to the resolution of the F480M filter. Source detection is carried out using SEXTRACTOR (E. Bertin & S. Arnouts 1996) through two parallel steps. We construct two inverse-variance-weighted detection images: one from the short-wavelength (SW; F090W, F115W, F150W, F200W) bands to preserve spatial resolution, and one from the long-wavelength (LW; F277W, F356W, F444W) bands to ensure sensitivity to red or dusty sources (e.g., LRDs). Unlike the science images, these have not been PSF-matched. These SW and LW detection catalogs are subsequently merged into a single, combined catalog. Photometry is then measured using PHOTUTILS (L. Bradley et al. 2020) in a range of circular apertures ($D = 0''.1$ – $1''.2$). Photometric uncertainties are estimated by placing random apertures in empty regions around each source. Unless stated otherwise, we adopt total fluxes measured within a $D = 0''.2$ aperture throughout this work.

2.2. NIRSpec Observations

GLIMPSE-D NIRSpec data were obtained during a campaign (DDT #9223; PIs: S. Fujimoto & R. Naidu) to study a promising Pop III galaxy candidate (S. Fujimoto et al. 2025a). GLIMPSE-D obtained a total of three G395M/F295LP MSA pointings, totaling 29.78 hr of total exposure time. The pointing center differed for each of the three configurations to maximize the total object yield. In total, the GLIMPSE-D sample contains 384 spectra, with depths varying from 9.2 hr to ~ 30 hr. All of the planned observations were successfully executed between 2025 June 30 and July 2. For each MSA configuration, GLIMPSE-D employed a standard three-point nod pattern at an aperture position angle $\text{PA } \vee 3 = 270^\circ.5$, using 494 groups per integration with the NRSIRS2 readout mode. The full details of the target selection, prioritization, and MSA planning will be presented in a forthcoming survey paper (S. Fujimoto et al. 2025b).

2.3. G395M Data Reduction and Calibration

The GLIMPSE-D MSA spectra were uniformly reduced using MSAEXP (v0.9.8; G. Brammer 2022). This procedure starts with the level-2 calibrated products obtained from MAST and applies a number of corrections that include 1/f

noise, artifact detection and removal, and a bias correction in individual exposures (see J. Rigby et al. 2023 for more details). Together with the JWST pipeline, MSAEXP sets the slit World Coordinate System, performs flat-fielding, and computes an initial pass-loss correction. Each one of the two-dimensional shutters is then drizzled onto a common pixel grid. Using the standard approach, the background subtraction is performed locally using stacked, source-free shutters. The one-dimensional spectra are then obtained via the optimal extraction method (e.g., P. Arrabal Haro et al. 2023; A. de Graaff et al. 2024), in which the center and width of the extraction “aperture” vary depending on the best-fit Gaussian model (K. Horne 1986), similar to the methodology widely adopted by a variety of other works (e.g., B. Wang et al. 2023; J. E. Greene et al. 2024; V. Kokorev et al. 2024b).

The absolute flux calibration of MSA spectra can be influenced by several factors, including the position of the source within the shutter, calibration and astrometric uncertainties, and the intrinsic morphology of the source. To correct for these effects and derive an overall slit-loss correction, we rescale the extracted one-dimensional spectra by convolving them with all available NIRCam filters and comparing the resulting flux densities to the total photometry from the GLIMPSE catalog. A wavelength-dependent correction is then obtained by fitting a second-order polynomial to these differences.

The target in this paper was observed in all three configurations; however, one was severely contaminated by sources in the same row, making the data recovery for that configuration unfeasible. Despite this, GLIMPSE-17775 is securely detected in two configurations (33,262 s and 40,703 s), yielding a total integration time of 20.55 hr. We combine the extracted and separately calibrated one-dimensional spectra using inverse-variance weighting. The stacked spectrum is shown in Figure 1. We also make the spectrum publicly available.¹⁷

3. Emission-line Analysis

3.1. Spectroscopic Redshift

The spectrum of GLIMPSE-17775 reveals a remarkably wide variety of significantly detected emission and absorption lines. Before focusing on individual features and their complex components, we perform an initial tally of all detectable lines using a heavily modified version of MSAEXP (G. Brammer 2022; V. Kokorev et al. 2024b). The key modifications allow varying the line widths and fitting multiple components, albeit tied to the same redshift.

We fit the full spectrum using MSAEXP, adopting Gaussian profiles for emission lines and a three-segment cubic spline to model the continuum. This minimal spline structure, appropriate given the high average S/N of GLIMPSE-17775 ($\gtrsim 30$), helps prevent overfitting, particularly of features that may be slightly offset in wavelength or velocity space from the average redshift. Emission-line positions are fixed; narrow components are allowed a FWHM between 150 and 800 km s^{−1}, and permitted transitions may include a broad component (800–5000 km s^{−1}).

This yields a redshift of $z_{\text{spec}} = 3.5010 \pm 0.0001$ (see Table 1). However, several strong lines exhibit significant pixel-level offsets of 100–200 km s^{−1} relative to this value. These discrepancies are statistically robust and appear in both

¹⁷ <https://zenodo.org/records/17575391>

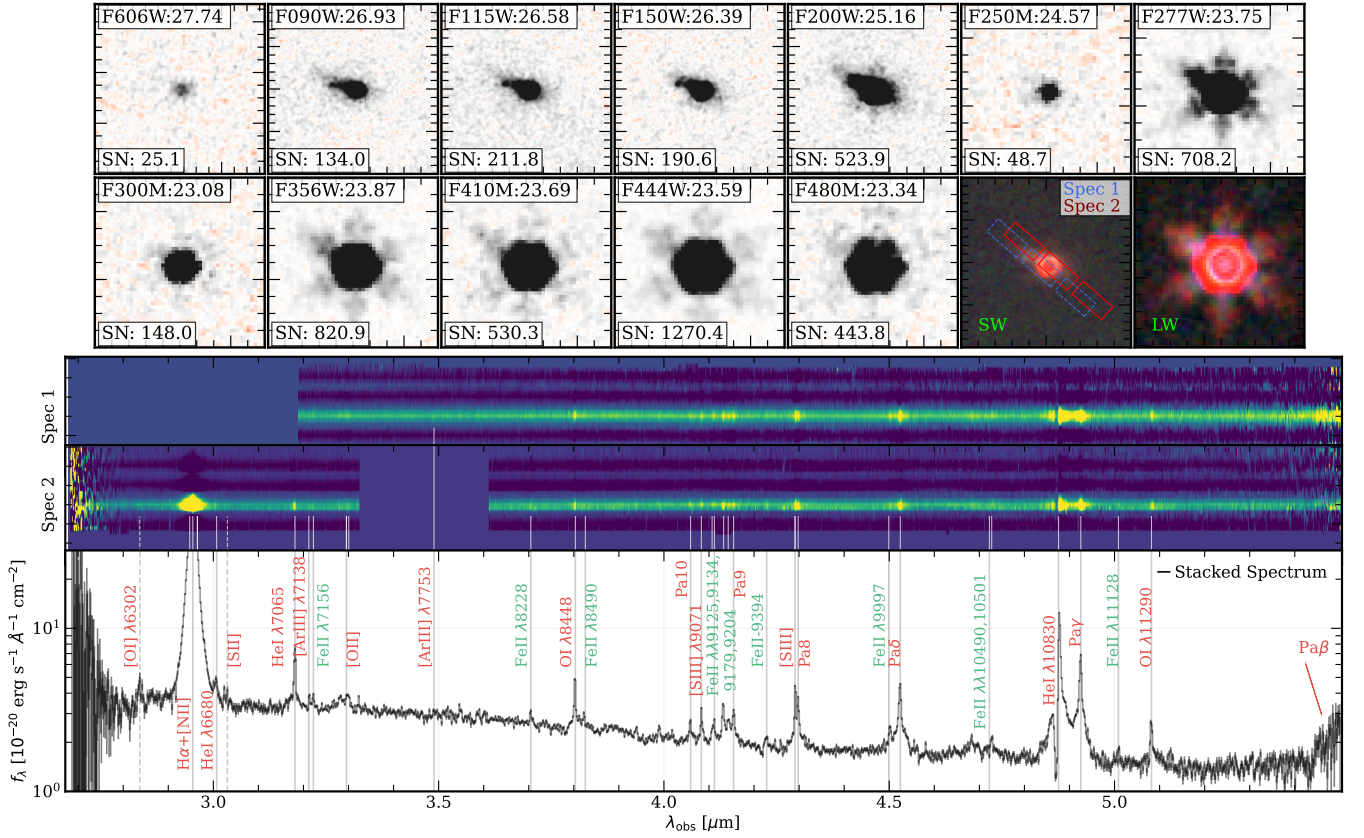


Figure 1. Top: JWST/NIRCam and HST 2'0 stamps and red, green, blue short-wavelength (SW) and long-wavelength (LW) color images, comprising the F115W, F150W, and F200W and F277W, F356W, and F444W bands, respectively. MSA shutters for both configurations covering GLIMPSE-17775 are shown in blue and red, respectively. The source morphology is resolved and extended up to $\sim 2 \mu\text{m}$, and then appears to transition to a more PSF-dominated and compact shape, echoing a growing sample of LRDs with extended rest-UV morphology (I. Juodžbalis et al. 2024; I. Labbe et al. 2024; J. Matthee et al. 2024; P. Rinaldi et al. 2025a). This likely hints at the presence of the host galaxy in the filters covering the rest-UV. In each panel, we show the total AB magnitude as presented in the GLIMPSE catalog (H. Atek et al. 2025; V. Kokorev et al. 2025). The source is exceptionally bright ($M_{444} \sim 23.6$) and is detected in most JWST bands at $>100\sigma$. Middle: two-dimensional MSA G395M spectra covering GLIMPSE-17775. Bottom: combined one-dimensional spectrum (see, e.g., V. Kokorev et al. 2023; A. de Graaff et al. 2025a for the extraction method) of the LRD in the observed frame. We show the data in black, and the uncertainty as a black shaded region. Fixing the systemic redshift to the [S III] $\lambda 9071$ line ($z_{\text{spec}} = 3.50102 \pm 0.00019$), we show the positions and label the prominent emission with significant ($\geq 3\sigma$) detections as solid vertical lines. Iron lines are shown separately in green. Due to the sheer number of features, not all could be labeled; we therefore display all features in Figure 2. Emission lines for which only upper limits are obtained are shown with dashed lines.

stacked and individual spectra, suggesting that the derived redshift reflects a weighted average dominated by high-S/N lines. To properly assess velocity structure, a consistent systemic redshift is required. In the following sections describing the detailed line analysis, we adopt the centroid of a narrow forbidden line as the systemic reference frame (specifically [S III] $\lambda 9071$; see Section 3.3).

Using the updated redshift alongside the strong-lensing model of AS1063 (A. Zitrin et al. 2015; L. Furtak et al. 2026, in preparation), we recalculate the lensing magnification to be $\mu = 2.04 \pm 0.21$, consistent with previous estimates based on z_{phot} . With redshift and magnification now fixed, we turn to a deeper analysis of the emission-line properties.

3.2. Line Identification

Beyond velocity offsets, the initial MSAEXP fit highlights several notable features. Broad hydrogen lines, ubiquitous in LRDs (e.g., D. D. Kocevski et al. 2023; V. Kokorev et al. 2023; L. J. Furtak et al. 2024; J. Matthee et al. 2024; A. de Graaff et al. 2025b; R. E. Hviding et al. 2025; R. P. Naidu et al. 2025; A. J. Taylor et al. 2025), are strongly favored across the Balmer and Paschen series, with $\Delta\chi^2 > 10$. These include H α and Pa10 through Pa γ . Although Pa β falls near the

edge of the detector, we nonetheless observe a prominent broad wing at its expected location (see Figure 1).

Broad components are also detected in several helium and oxygen lines: He I $\lambda 6680$, He I $\lambda 7065$, O I $\lambda 8448$, He I $\lambda 10830$, and O I $\lambda 11290$. Notably, the He I $\lambda 10830$ line shows a classic P-Cygni profile with a clear blueshifted absorption trough. While Balmer absorption is becoming a familiar sight in LRDs (R. Maiolino et al. 2024; J. Matthee et al. 2024; A. de Graaff et al. 2025b; D. D. Kocevski et al. 2025; R. P. Naidu et al. 2025), helium absorption remains rare, with only a few published cases at high z (e.g., I. Juodžbalis et al. 2024; R. P. Naidu et al. 2024; B. Wang et al. 2025). Curiously, this was also recently reported in local analogs of LRDs (X. Lin et al. 2026).

The most striking feature is the detection of an extensive Fe II emission forest in the rest-NIR—at least 16 distinct features are identified by MSAEXP and highlighted in green in Figure 1. Previous detections of permitted Fe II in LRDs have been mostly limited to rest-UV/optical wavelengths using low-resolution PRISM spectra (R. Tripodi et al. 2025), with only few examples in the NIR (e.g., see broad Fe II $\lambda 9200$ I. Labbe et al. 2024). Iron has also been identified in local LRD analogs (X. Lin et al. 2026; X. Ji et al. 2026). Only recently

Table 1
Source Properties

Parameter ^a	GLIMPSE-17775
ID	17775
R.A. [deg]	342.20080
Decl. [deg]	-44.54366
z_{phot} (EAZY)	4.2 ± 0.1
z_{spec} ([S III] $\lambda 9071$)	3.50102 ± 0.00019
μ	2.04 ± 0.21
M_{UV} [AB mag]	-17.27 ± 0.05
$r_{\text{eff,UV}}$ [pc]	1000 ± 200
$r_{\text{eff,opt}}$ [pc]	< 300
β	-0.69 ± 0.12
$\log_{10}(M_{\text{BH}}/M_{\odot})$	6.65 ± 0.15
L_{bol} [erg s ⁻¹]	$(1.06 \pm 0.14) \times 10^{45}$
λ_{edd}	1.86 ± 0.25
$\log_{10}(M_{*}/M_{\odot})_{M_{\text{UV}}}$	< 7.5
A_{V}	0.1 ± 0.3
$f_{\nu,4050\text{\AA}}/f_{\nu,3670\text{\AA}}$	2.02 ± 0.10

Note.^a Physical parameters are corrected for the lensing magnification.

have a few Fe II emitters been reported in medium-resolution (F. D’Eugenio et al. 2025) and high-resolution spectra (A. Torralba et al. 2026). The richness of these lines in GLIMPSE-17775 hints at a dense, partially shielded gas phase, where processes such as Ly β fluorescence and continuum pumping may be enhancing Fe II emission (T. A. A. Sigut & A. K. Pradhan 1998, 2003; T. A. A. Sigut et al. 2004), offering a new diagnostic window into early AGN environments.

In summary, GLIMPSE-17775 exhibits an exceptionally rich and kinematically complex array of emission and absorption features. While the initial MSAEXP modeling provides a solid foundation for redshift estimation and line identification, it lacks the flexibility to capture the full diversity of line profiles in this high-S/N dataset. In the following sections, we peel back the spectral layers with increasing precision.

3.3. Line Fitting

The exceptional depth of the G395M spectrum for GLIMPSE-17775 enables detailed modeling of both multiple kinematic components and potential velocity offsets between line species. Unless noted otherwise, each line is fit with a single Gaussian narrow component (FWHM = 100–800 km s⁻¹). For permitted lines flagged by MSAEXP as potentially broad, we follow the approach of V. Rusakov et al. (2026) and model the profile as an intrinsic Gaussian core broadened by electron scattering, implemented via convolution with an exponential-wing component. The intrinsic Gaussian FWHM is allowed to vary between 500 and 5000 km s⁻¹. The characteristic scattering width (i.e., the strength of the exponential wings, W) is treated as a free parameter and is allowed to vary independently between different transitions. For each broad line, we also perform an alternative fit with the W parameter fixed to zero (pure Gaussian) and compare the resulting goodness of fit to assess whether the exponential wings are statistically required. Line centers are allowed to vary independently unless otherwise specified.

Absorption lines are modeled using a Gaussian optical-depth profile, where the observed flux is $F_{\text{obs}} = F_{\text{em}} e^{-\tau(\lambda)}$, where $\tau(\lambda) = \tau_0 \exp[-(\lambda - \lambda_{\text{abs}})^2 / 2\sigma_{\text{abs}}^2]$. The free parameters are then the central optical depth τ_0 , velocity width

Δv_{abs} , and redshift z_{abs} (see I. Juodžbalis et al. 2024). Widths and redshifts (for both narrow and broad components) are typically free, with redshift limited to ± 0.1 from the MSAEXP value. Local continua are fit using first-order polynomials.

All individual model components are first initialized and coadded on an oversampled wavelength grid. To take into account the wavelength-dependent resolution of the grating, we interpolate our model onto a variable step grid while making sure that the total integrated flux is preserved. Further, we increase the nominal spectral resolution by a factor of 1.7, as it has been shown that the spectral resolution for a pointlike source falling within a shutter is higher than that of a uniformly illuminated slitlet (A. de Graaff et al. 2024). Fitting uses nonlinear χ^2 minimization, with uncertainties derived from multivariate resampling of the covariance matrix.

Given the complexity, the spectrum is divided into six windows grouped by line species or proximity (see Figure 2). Below, we describe the assumptions adopted for each window. Final fluxes and equivalent widths of the narrow and broad lines are listed in Tables 2 and 3, respectively, while kinematics are reported in Tables 4 and 5. We show detailed line fits in Figure 2.

3.3.1. H α Complex

H α is modeled with independent narrow, broad, and absorption components. [O I] $\lambda 6302$ and He I $\lambda 6680$ are modeled with narrow and broad profiles. [N II] lines are fixed at a 1:3 ratio and share kinematics; [S II] lines have independent amplitudes but tied velocities and widths.

We find that broad H α and He I $\lambda 6680$ are strongly favored to have exponential wings ($\Delta\text{BIC} \sim -500$), yielding a FWHM (of the intrinsic Gaussian core) of ~ 1000 km s⁻¹—significantly narrower than the ~ 3000 km s⁻¹ derived from pure Gaussians. Although no distinct H α absorption component is explicitly detected, likely due to insufficient resolution, the overall asymmetry of the line is best reproduced when a weak, blueshifted absorber is included in the fit. This may indicate subtle self-absorption or partial obscuration within the dense cocoon. We return to the implications of this profile shape later.

3.3.2. He I $\lambda 7065$

The He I $\lambda 7065$ line shows a prominent blueshifted absorption component. We fit it with narrow, broad, and absorption profiles. The rest of the lines, including [Ar III] $\lambda 7138$, Fe II $\lambda 7156$, and the [O II] triplet ($\lambda\lambda 7323, 7325, 7332$), are modeled as independent narrow lines.

He I $\lambda 7065$ shows a strong preference for an exponential profile ($\Delta\text{BIC} \sim -16$) with a FWHM = 473 ± 116 km s⁻¹. A residual bump spanning from $\sim \lambda 7290$ to $\lambda 7320$ is likely a combination of multiple Fe II, such as $\lambda 7290, \lambda 7308$, and potentially Fe III $\lambda 7319.65$ lines. Curiously, the latter line is often labeled as “hazy” in various emission-line libraries (e.g., A. Kramida et al. 2024), reflecting its undefined shape as a result of pressure broadening or scattering in very dense gas.

3.3.3. O I $\lambda 8448$ and Iron Lines

In this part of the spectrum the O I $\lambda 8448$ is fit with narrow and broad components. All of the iron lines, including Fe II $\lambda\lambda 8228, 8239, 8289, 8470, 8490$ as well as a potential Fe III $\lambda 8306$, are fit as narrow lines with shared kinematics. O I components are fit independently.

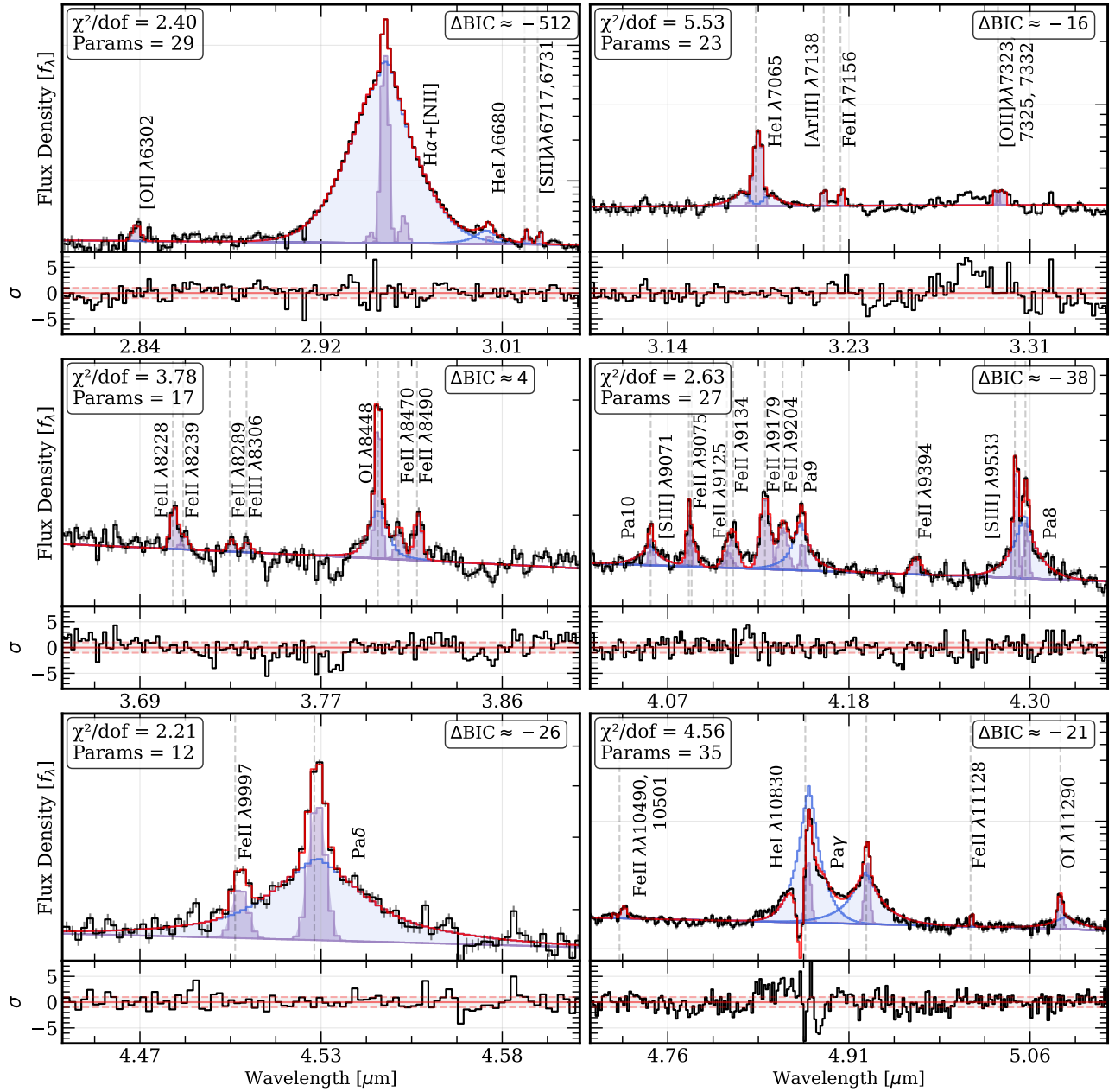


Figure 2. A staggering abundance of spectral lines in GLIMPSE-17775 at $z = 3.501$. For each spectral window defined in Section 3.3, we show the data (black), the best-fit narrow and broad components (dark purple and blue), and the total best-fit model including the continuum (red). All broad components were fit with models allowing exponential wings. The ΔBIC between exponential and Gaussian fits is reported in the top right of each panel; negative values indicate a preference for an exponential profile. The lower panels display the uncertainty-weighted residuals for each fit.

We find that the broad O I component is best described by a standard Gaussian profile, with a stronger statistical preference over exponential wings ($\Delta\text{BIC} \sim 4$). However, we caution that a trough at $\sim 3.77 \mu\text{m}$, likely an artifact, may affect the reliability of the fit in this region.

3.3.4. Iron Forest and Paschen Lines

This spectral region is among the most complex, due to the dense clustering of emission features. The Paschen lines Pa10–Pa8 are modeled with narrow and broad components. To reduce the number of free parameters, we tie all narrow Paschen components together kinematically, and do the same for the broad components. Fe II $\lambda\lambda 9075, 9125, 9134,$

9179, 9204, 9394 are fit with a shared centroid and width. [S III] lines are fit independently.

The [S III] $\lambda 9071$ line is used to define the systemic redshift: $z = 3.50102 \pm 0.00019$. This is done because forbidden lines, such as [S III] or [O III], have lower critical densities that are incompatible with the dense-gas envelopes giving rise to the other lines we observe. These lines may instead come from the host galaxy itself (e.g., R. Maiolino et al. 2024), and we therefore choose this line to define the overall redshift of the system and to measure all the offsets relative to it. All velocity offsets discussed in the subsequent sections are computed relative to this reference frame. The Paschen lines show strong preference for exponential wings

Table 2Fluxes of Narrow Emission Lines and Their Rest-frame Equivalent Widths^a

Line	λ_{rest} (Å)	Flux (10^{-20} erg s $^{-1}$ cm $^{-2}$)	EW ₀ (Å)
Narrow Emission Lines			
O I	6302.0	54.0 ± 21.2	3.3 ± 1.2
[N II]	6549.0	25.6 ± 10.3	1.6 ± 0.7
H α	6562.8	2622.9 ± 200.7	167.7 ± 12.8
[N II]	6584.0	77.1 ± 37.1	4.9 ± 1.6
He I	6680.0	14.7 ± 11.1	0.8 ± 0.7
[S II]	6717.0	22.2 ± 0.1	1.5 ± 0.7
[S II]	6731.0	21.4 ± 0.1	1.4 ± 0.5
He I	7065.0	461.7 ± 210.2	31.0 ± 12.2
[Ar III]	7138.0	12.6 ± 3.4	1.0 ± 0.3
Fe II	7156.0	26.2 ± 7.2	1.8 ± 0.2
[O II]	7323.0	26.0 ± 23.0	1.7 ± 1.5
[O II]	7325.0	16.9 ± 1.8	1.2 ± 0.1
[O II]	7332.0	26.2 ± 9.1	1.8 ± 0.6
Fe II	8228.0	23.7 ± 4.2	2.0 ± 0.3
Fe II	8239.0	7.2 ± 4.0	0.6 ± 0.3
Fe II	8289.0	6.3 ± 3.7	0.5 ± 0.3
Fe III	8306.0	5.7 ± 3.6	0.5 ± 0.3
O I	8448.0	67.6 ± 2.4	6.1 ± 1.3
Fe II	8470.0	16.2 ± 5.0	1.4 ± 0.4
Fe II	8490.0	26.7 ± 4.3	2.4 ± 0.4
Pa10	9015.0	16.5 ± 5.0	1.8 ± 0.5
[S III]	9071.0	27.2 ± 2.3	3.5 ± 0.3
Fe II	9075.0	30.0 ± 4.7	3.4 ± 0.5
Fe II	9125.0	16.8 ± 5.2	1.9 ± 0.6
Fe II	9134.0	36.9 ± 5.0	4.1 ± 0.6
Fe II	9179.0	95.0 ± 5.5	10.6 ± 0.6
Fe II	9204.0	50.1 ± 6.1	5.6 ± 0.7
Pa9	9229.0	19.4 ± 7.0	2.2 ± 0.8
Fe II	9394.0	17.3 ± 3.9	2.0 ± 0.5
[S III]	9533.0	84.7 ± 5.9	10.2 ± 0.7
Pa8	9545.0	44.8 ± 11.3	5.4 ± 1.4
Fe II	9997.0	29.5 ± 4.5	3.8 ± 0.6
Pa δ	10049.0	104.2 ± 12.2	13.7 ± 1.7
Fe II	10490.0	8.0 ± 2.7	1.1 ± 0.4
Fe II	10501.0	24.4 ± 5.9	3.1 ± 0.8
He I	10830.0	163.0 ± 96.0	22.6 ± 13.3
Pa γ	10938.0	146.3 ± 25.5	20.9 ± 3.8
Fe II	11128.0	12.3 ± 3.7	1.8 ± 0.6
O I	11290.0	49.2 ± 4.1	7.7 ± 2.3

Note.^a Not corrected for lensing magnification.

($\Delta\text{BIC} \sim -38$), with a FWHM = 766 ± 202 km s $^{-1}$ —narrower than H α , but consistent within 2σ .

3.3.5. Pa δ

The Fe II $\lambda 9997$ and Pa δ complex is modeled with an independent narrow component for Fe II and both narrow and broad components for Pa δ . There is no evidence for absorption. Exponential wings are again preferred ($\Delta\text{BIC} \sim -26$). The resulting FWHM of 1672 ± 458 km s $^{-1}$ is notably larger than those of both higher-order Paschen lines and H α , though still consistent with the latter within 2σ . It is also possible that the line shape is impacted by Fe II $\lambda 9997$ and the undetected (but likely present) Fe II $\lambda\lambda 10131, 10173$ lines.

Table 3Fluxes of Broad Emission Lines and Their Rest-frame Equivalent Widths^a

Line	λ_{rest} (Å)	Flux (10^{-20} erg s $^{-1}$ cm $^{-2}$)	EW ₀ (Å)
Broad Emission Lines			
H α	6562.8	14803.5 ± 198.6	946.3 ± 15.6
He I	6680.0	144.6 ± 44.7	9.4 ± 2.9
He I	7065.0	824.2 ± 210.0	55.4 ± 15.1
Pa10	9015.0	53.7 ± 11.1	5.8 ± 1.3
Pa9	9229.0	119.0 ± 13.5	13.6 ± 1.5
Pa8	9545.0	191.2 ± 17.0	23.1 ± 2.1
Pa δ	10049.0	370.6 ± 15.0	48.8 ± 2.3
He I	10830.0	2294.9 ± 212.6	318.8 ± 30.3
Pa γ	10938.0	652.3 ± 14.0	92.9 ± 2.4
O I	8448.0	93.7 ± 7.9	8.4 ± 0.7
O I	11290.0	106.2 ± 63.9	16.6 ± 10.4

Note.^a Not corrected for lensing magnification.**Table 4**

Kinematic Properties of Narrow Lines

Line	λ_{rest} (Å)	FWHM (km s $^{-1}$)	Δv (km s $^{-1}$)
Narrow Emission Lines			
[O I]	6300.3	455 ± 175	−101 ± 81
H α	6562.8	304 ± 8	+145 ± 13
[N II]	6583.4	347 ± 117	+84 ± 47
[S II]	6716.4	232 ± 62	+82 ± 33
[S II]	6731	232 ± 62	+82 ± 33
He I	6678.2	87 ± 14	+40 ± 14
He I	7065.2	408 ± 644	+112 ± 396
[Ar III]	7135.8	89 ± 50	−26 ± 20
Fe II	7155.2	96 ± 50	+15 ± 15
[O II]	7320.0	400 ± 120	−37 ± 72
Fe II	8228.0	290 ± 48	+21 ± 5
Fe II	8239.0	290 ± 48	+21 ± 5
Fe II	8289.0	290 ± 48	+21 ± 5
Fe III	8306.0	290 ± 48	+21 ± 5
O I	8446.4	235 ± 30	−43 ± 19
Fe II	8470.0	290 ± 48	+21 ± 5
Fe II	8490.0	290 ± 48	+21 ± 5
Pa10	9014.9	259 ± 36	−46 ± 19
[S III]	9071.0	230 ± 13	...
Fe II	9075.0	486 ± 24	−21 ± 17
Fe II	9125.0	486 ± 24	−21 ± 17
Fe II	9134.0	486 ± 24	−21 ± 17
Fe II	9179.0	486 ± 24	−21 ± 17
Fe II	9204.0	486 ± 24	−21 ± 17
Pa9	9229.0	259 ± 35	−46 ± 19
Fe II	9394.0	486 ± 24	−21 ± 17
Pa8	9545.6	259 ± 36	−46 ± 19
[S III]	9533	230 ± 13	+23 ± 13
Fe II	9997.0	331 ± 55	+90 ± 25
Pa δ	10049.4	332 ± 22	+21 ± 15
He I	10830.3	287 ± 46	+112 ± 26
Pa γ	10938.1	267 ± 25	+26 ± 17
Fe II	10490.0	332 ± 96	+44 ± 43
Fe II	10501.0	332 ± 96	+44 ± 43
Fe II	11128.0	153 ± 116	+37 ± 30
O I	11290.0	260 ± 58	−33 ± 24

Note. Velocity offsets (Δv) calculated relative to [S III] $\lambda 9071$ at $z = 3.50102$.

Table 5
Kinematic Properties of Broad and Absorption Lines

Line	λ_{rest} (Å)	FWHM (km s ⁻¹)	W (km s ⁻¹)	Δv (km s ⁻¹)
Broad Emission Lines				
H α	6562.8	1024 $\pm$ 21	3010 $\pm$ 450	-7 $\pm$ 16
He I	6678.2	1041 $\pm$ 792	3890 $\pm$ 2083	-137 $\pm$ 151
He I	7065.2	473 $\pm$ 116	1000 $\pm$ 600	-136 $\pm$ 82
O I	8446.4	562 $\pm$ 331	1120 $\pm$ 590	+39 $\pm$ 80
Pa10	9014.9	766 $\pm$ 202	2100 $\pm$ 600	-155 $\pm$ 39
Pa9	9229.0	766 $\pm$ 202	2100 $\pm$ 600	-155 $\pm$ 39
Pa8	9545.6	766 $\pm$ 202	2100 $\pm$ 600	-155 $\pm$ 39
Pa δ	10049.4	1672 $\pm$ 458	3500 $\pm$ 400	+23 $\pm$ 44
He I	10830.3	536 $\pm$ 45	2760 $\pm$ 220	+154 $\pm$ 22
Pa γ	10938.1	1035 $\pm$ 322	2400 $\pm$ 260	-75 $\pm$ 31
O I	11287.0	1188 $\pm$ 1099	4300 $\pm$ 2000	+245 $\pm$ 214
Absorption Lines				
H α	6562.8	1000 $\pm$ 123	...	-200 $\pm$ 32
He I	7065.2	587 $\pm$ 1024	...	-52 $\pm$ 83
He I	10830.3	794 $\pm$ 21	...	+45 $\pm$ 22

Note. Velocity offsets (Δv) calculated relative to [S III] λ 9071 at $z = 3.50102$.

3.3.6. He I λ 10830 and Pa γ

The final window contains a complicated blend of broad He I λ 10830 and Pa γ lines, with a clear blueshifted absorption in the former. We fit three components (narrow, broad, and absorption) to He I λ 10830 and narrow plus broad components to Pa γ , keeping everything kinematically independent. We fit narrow Fe II lines at 10490, 10501, and 11128 Å with fixed redshift and line width. Finally, O I λ 11290 is fit with independent narrow and broad components. Exponential wings are again preferred over a pure Gaussian profile.

4. Data Analysis

4.1. Morphology

LRDs are unresolved in the rest-optical by definition, with measured sizes consistent with the PSF HWHM ($\lesssim 0''.07$ in F444W), and in some cases even smaller when dithers align favorably (e.g., I. Labbé et al. 2025). Gravitational lensing can further push constraints on their intrinsic sizes to $\lesssim 100$ pc (L. J. Furtak et al. 2024).

In the rest-UV, however, a more complex picture is emerging. Several studies have now reported faint, extended, asymmetric components adjacent to the compact core (V. Kokorev et al. 2024b; I. Labbé et al. 2024; J. Matthee et al. 2024; P. Rinaldi et al. 2025a, 2025b), often suppressed by surface-brightness dimming. As shown in Figure 1, GLIMPSE-17775 likewise consists of two components out to F200W (rest ~ 4000 Å), coincident with the Balmer break. However, we note that the break itself is much weaker than found in objects with very similar spectra (e.g., B. Wang et al. 2024; see Figure 3). At longer wavelengths, a point-source (PS) morphology dominates, consistent with a black-hole-dominated core. To quantify this transition, we model the NIRCcam imaging using PYSERVIC (I. Pasha & T. B. Miller 2023) with a minimal configuration: a fixed-center PS plus a freely offset Sérsic profile. Normalizations, n , and r_{eff} are all

allowed to vary, and uncertainties are drawn from the Markov Chain Monte Carlo posteriors (e.g., V. Kokorev et al. 2024a).

Our fits show that the host galaxy dominates up to $\lambda_{\text{rest}} \sim 4000$ Å, where the host and nucleus contribute roughly equally, beyond which the PS prevails to $\lambda_{\text{rest}} \sim 1 \mu\text{m}$. Because the NIRCcam LW detector has $\gtrsim 2\times$ poorer resolution than the SW, a clean separation of host and nucleus in the rest-optical is impractical (see, e.g., K. E. Whalen et al. 2026). Fortunately, F200W lies at the transition where both the resolution and flux ratio are favorable, so we use that band to illustrate our galaxy/LRD decomposition. We illustrate this in Figure 4, which also shows the increasing PS fraction with wavelength, while acknowledging the uncertainties at longer wavelengths.

After applying the lensing correction, we find that the extended component dominating the UV light has a radius of $r_{\text{eff}} \sim 1000$ pc, whereas the actual compact LRD, dominant in the rest-optical, is fully consistent with the PSF and $r_{\text{eff}} < 300$ pc.

4.2. Dust Attenuation

The ratio between observed emission-line fluxes is commonly used to estimate dust extinction. For LRDs, this has typically been achieved through the Balmer decrement, provided that multiple Balmer lines are available. In our case, only H α is detected. While the Paschen series, of which five lines are observed, could, in principle, be used to estimate extinction, the likely presence of stratified dense gas in LRDs (A. de Graaff et al. 2025b; R. P. Naidu et al. 2025) implies that the intrinsic line ratios may deviate significantly from standard Case B recombination expectations (S.-J. Chang et al. 2026). Further, at the wavelengths of the Paschen series, the attenuation curves are practically flat (e.g., D. Calzetti et al. 2000). Small offsets of the Case B ratios would have a significant effect on the derived dust attenuation, but in return require extreme precision on measured line fluxes to be meaningful. Our uncertainties on, e.g., Pa γ /Pa δ are small, roughly 10%–15%, yet this still leads to an impractically high uncertainty on A_V . To circumvent this limitation, we instead employ the ratio between the permitted O I λ 8448 and O I λ 11290 lines.

These transitions share a common energy level, and under the assumption of Bowen (Ly β) fluorescence, all transitions through the λ 11290 transition also cascade through the λ 8448. This makes their intrinsic intensity set solely by the ratio of the inverse of their wavelengths.

To ensure we are comparing the same physical gas component, we isolate the narrow emission peaks in both transitions, which show consistent widths of FWHM ~ 250 km s⁻¹. This similarity supports the use of their flux ratio as a dust probe. We adopt the Small Magellanic Cloud (SMC) attenuation curve (K. D. Gordon et al. 2003), widely used for high-redshift galaxies and reddened AGN (P. F. Hopkins et al. 2004; P. L. Capak et al. 2015; N. A. Reddy et al. 2015, 2018; V. Kokorev et al. 2023; A. J. Taylor et al. 2025). Assuming an intrinsic ratio of $(8448/11290)_{\text{int}} = 1.336$ (D. E. Osterbrock 1989), we infer $A_V = 0.1 \pm 0.3$ mag, consistent with negligible extinction.

We caution, however, that this ratio probes only the narrow-line-emitting gas, which could originate either in the host galaxy or within a narrow-line region associated with the AGN. If the latter is the case, then the red optical continuum slope ($\beta_{\text{opt}} \sim 0.35$) would suggest that the observed SED is intrinsically red rather than reddened by dust. If instead the

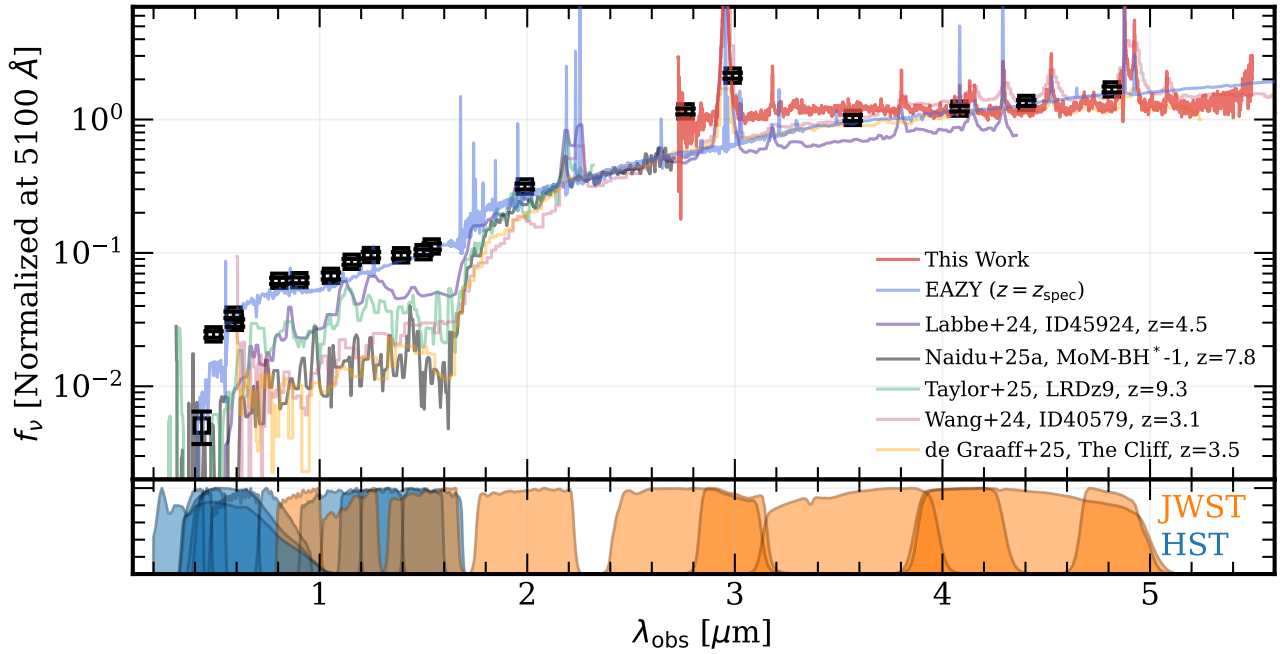


Figure 3. The diversity of Balmer breaks in LRDs. The black points show the HST and JWST GLIMPSE photometry of GLIMPSE-17775. The blue line shows best-fit EAZY SED fit to the photometry only, fixing the redshift to the z_{spec} . The maroon line shows the combined and photometry-corrected G395M spectrum. While the red color in F200W–F277W is partially influenced by a bright $\text{H}\alpha$ line, the Balmer break between F150W and F200W is still prominent. We further show spectra of various other LRDs (I. Labbe et al. 2024; B. Wang et al. 2024; A. de Graaff et al. 2025b; R. P. Naidu et al. 2025; A. J. Taylor et al. 2025), all shifted to $z = 3.501$ and normalized at 5100 Å. Finally, we show HST (blue) and JWST (orange) filter transmission curves below.

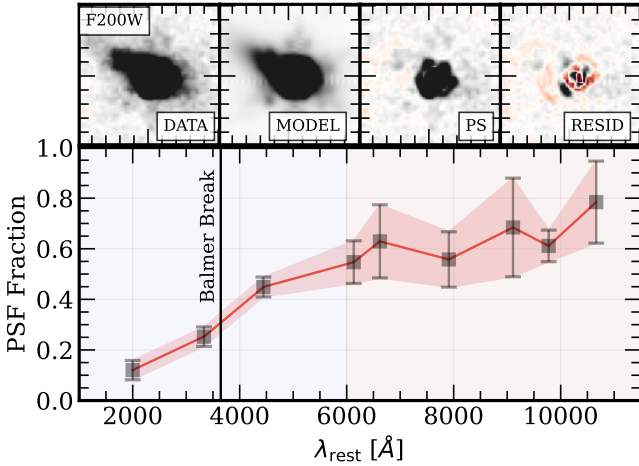


Figure 4. Top: two-component Sérsic+point-source (PS) fit to the F200W morphology of GLIMPSE-17775. Panels show, from left to right, the data, best-fit model, model with the extended component removed, and residuals. Bottom: fractional PS contribution vs. rest-frame wavelength. Shaded regions mark SW (blue) and LW (red) detectors, where the lower resolution of LW filters hinders reliable two-component decomposition.

narrow lines arise predominantly in the host galaxy, this constraint does not directly inform the origin of the continuum emission. Similar Case B-consistent narrow-line ratios have been reported in other LRDs (e.g., M. Brooks et al. 2025; G. P. Nikopoulos et al. 2025), where the broad lines were found to show significant departures from Case B.

Although the wavelength coverage of the red grating does not allow a Balmer decrement test for the broad-line emission (only $\text{H}\alpha$ is detected), we can perform a consistency check using the brightest broad Paschen lines ($\text{Pa}\gamma$, $\text{Pa}\delta$, and $\text{Pa}8$). Adopting intrinsic Case B ratios from P. J. Storey & D. G. Hummer (1995; $\text{Pa}\gamma/\text{Pa}\delta \simeq 1.5$ and $\text{Pa}\delta/\text{Pa}8 \simeq 1.8$), and comparing these to

our measured values of $\text{Pa}\gamma/\text{Pa}\delta = 1.77 \pm 0.15$ and $\text{Pa}\delta/\text{Pa}8 = 1.94 \pm 0.20$, we find the broad Paschen lines to be consistent with Case B within $\simeq 2\sigma$, particularly when systematic uncertainties from blending and profile decomposition are taken into account. If Case B is applicable in the dense environments surrounding the central AGN, this would be consistent with a low-dust or dust-free origin for the broad-line-emitting gas.

5. Line Properties

Our fitting procedure yields over 40 emission and absorption features, most of which are detected at a high ($\text{S/N} > 3$) significance. With all the pieces in place, we now comment on the line profiles, kinematics, and the specific line species that we identify.

5.1. Exponential Wings

As already noted throughout Section 4, and further highlighted in Figure 2, all permitted lines in the spectrum, with the exception of $\text{O I } \lambda 8448$, which is likely impacted by data-quality issues, are better fit when the broad component is convolved with an exponential profile. The statistical preference for this model (as measured by ΔBIC) is especially strong for our brightest line, $\text{H}\alpha$. To further demonstrate this, and to compare this fit with a more standard, Gaussian-only approach, we show both models in Figure 5. When the broad line is fit using only a Gaussian profile, large portions of the line are underfit at a $\gtrsim 3\sigma$ significance level, whereas a model that includes exponential wings shows a much smoother, albeit not perfect, residual plot.

This strong preference for exponential wings is consistent with expectations from electron scattering in dense, ionized gas. Unlike Gaussian profiles, which arise from Doppler broadening due to thermal and bulk motions around the

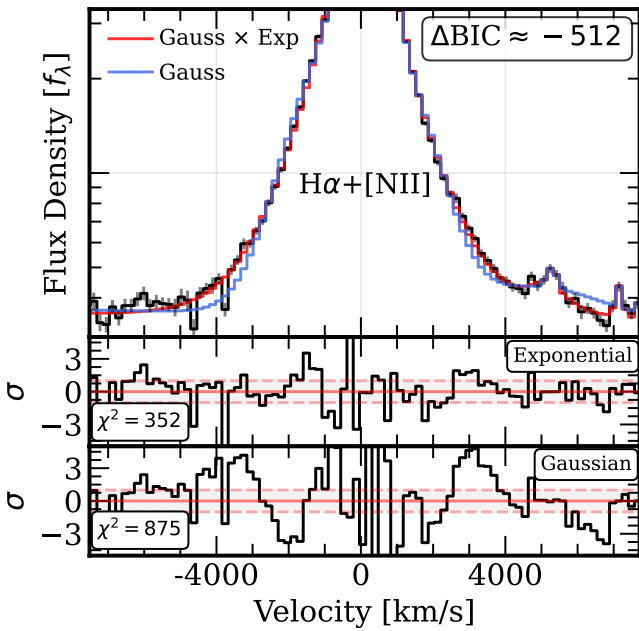


Figure 5. Exponential wings are required. Comparison of a Gaussian $\times$ exponential model (red) and a single Gaussian profile (blue) for H α . Uncertainties are shown as vertical lines in each spectral bin, although they are too small to be visible. The exponential model provides a far superior fit, with smoother residuals and a strongly preferred ΔBIC , highlighting the necessity of exponential wings to capture the broad-line shape.

supermassive black hole, exponential profiles arise in environments dominated by electron scattering (e.g., S.-J. Chang et al. 2026; V. Rusakov et al. 2026). The absence of strong broadening in the forbidden lines (e.g., [S II], [S III], and [N II]) already suggests that the densities where these electron-scattered profiles originate must already exceed multiple times the critical densities of these transitions ($n_e > 10^6$). Combined with the fact that these broad profiles are non-Gaussian in shape, both the likely volume and column densities are likely even higher, at $n_e = 10^8 \text{ cm}^{-3}$ and $N_e \sim 10^{24} \text{ cm}^{-2}$, respectively, as suggested by both K. Inayoshi & R. Maiolino (2025) and V. Rusakov et al. (2026).

Recent work (e.g., M. Brazzini et al. 2025) argues that, in classical broad-line region (BLR) environments, an electron-scattering origin for broad wings requires similar intrinsic and scattering widths across recombination lines. We therefore examine these criteria for the hydrogen transitions and find that, within the uncertainties of our fits, both the intrinsic and exponential widths are broadly comparable among the recombination lines. However, unlike classical BLRs, LRDs are expected to be strongly stratified, with different recombination transitions forming at different depths and probing different effective electron columns (e.g., A. de Graaff et al. 2025d; A. Sneppen et al. 2026). In such a structure, strict equality of widths is not expected even if a common scattering kernel shapes the wings. The modest variations observed here are therefore consistent with electron scattering in a dense, stratified cocoon.

It is important to note that such detailed profile decomposition is only possible due to the exceptionally high S/N per pixel achieved in the line wings of our NIRSpect G395M spectrum. As emphasized in V. Rusakov et al. (2026), these features would be impossible to distinguish with shallower

data or lower-resolution modes such as NIRSpect/PRISM, or even G395M exposures lacking comparable depth.

5.2. Line Profiles

Further, in Figure 6, we compare the best-fit models for all broad lines. As noted previously, with the exception of O I $\lambda 8448$, every broad line is better described by exponential wings, resulting in very similar overall line shapes. The main differences arise in their widths.

Despite some fits having sizable uncertainties, a trend emerges across line species. Hydrogen recombination lines (with the exception of Pa δ) and O I transitions consistently show comparable intrinsic FWHM values of $\sim 800 \text{ km s}^{-1}$. This agreement is not coincidental. Neutral oxygen and ionized hydrogen have nearly identical ionization potentials, enabling rapid charge-exchange coupling between $\text{O}^+ + \text{H}^0 \rightleftharpoons \text{O}^0 + \text{H}^+$ (D. E. Osterbrock & G. J. Ferland 2006). This continual electron exchange tightly locks the spatial distribution, ionization state, and kinematics of neutral oxygen to ionized hydrogen (and vice versa), so their broad-line profiles are naturally expected to track one another. The observed similarity therefore provides an important consistency check, reinforcing that the O I emission originates in the same dense, ionized gas as the hydrogen recombination lines. In contrast, the Fe II lines exhibit significantly narrower widths, matching the narrow cores of the permitted transitions rather than their exponential wings. This implies that the Fe II emission arises in a cooler, less turbulent zone exterior to the scattering-dominated region, but still closely coupled to the AGN continuum source.

By contrast, He I emission is systematically narrower. He I lines such as $\lambda 7065$ and $\lambda 10830$ do not couple to hydrogen via charge exchange, and their lower metastable level (2^3S) has distinct population physics, being fed by both collisional and recombination pathways. Further, one has to take into account the radiative-transfer effects, which make it possible to radiatively excite metastable ground-state electrons. As a result, He I emission may arise from a somewhat different spatial or kinematic region than H and O. We return to this point in more detail below.

Taken together, the similar FWHM of hydrogen and oxygen broad-line profiles, combined with the near-universal exponential wings seen across all permitted transitions, strongly suggests that the line shapes are set by a common, line-independent scattering kernel rather than by transition-specific processes. Electron (Thomson) scattering in a dense, ionized cocoon provides a natural explanation: It produces exponential wings of nearly identical form across species, with widths set primarily by the electron temperature and column density (typically $\sim 10^{24} \text{ cm}^{-2}$), and leaves only secondary variations from species-specific excitation or optical-depth effects (e.g., S.-J. Chang et al. 2026; V. Rusakov et al. 2026). In this framework, the narrower He I lines reflect stratification within the cocoon, where He I emission arises from an outer region that has a lower density, temperature, or column density, while the bulk hydrogen and oxygen emission share a common kinematic imprint within an inner, denser region.

5.3. Line Kinematics

Before we proceed to discussing the individual features in more detail, we would like to comment on the systematic

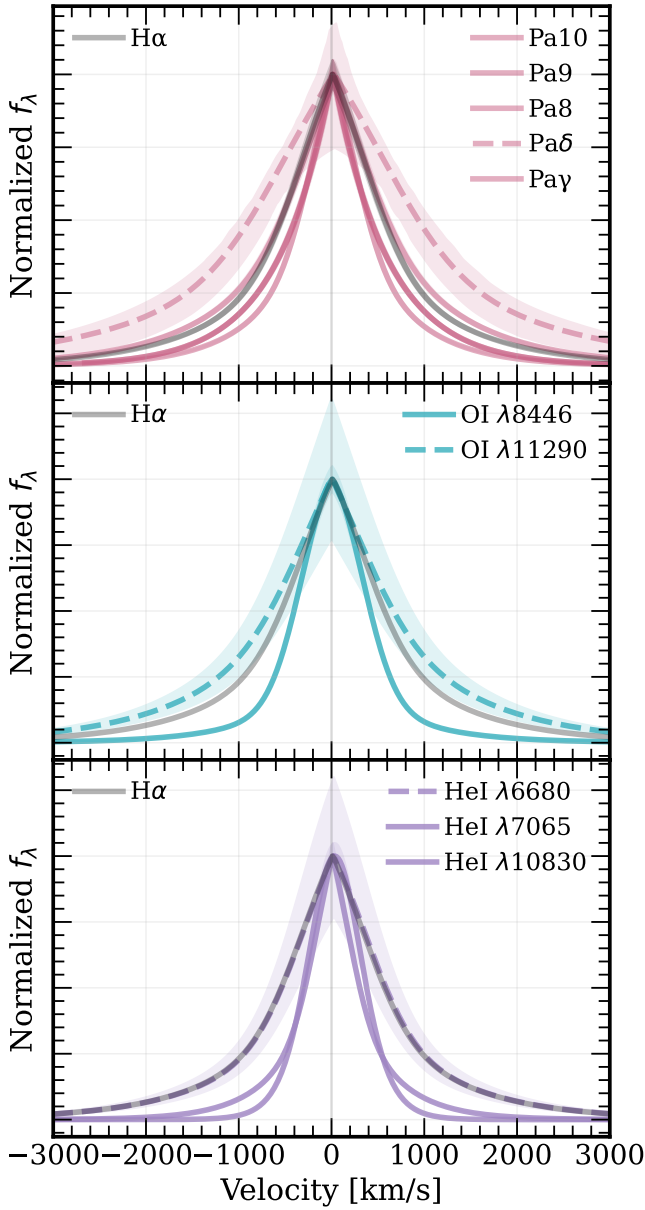


Figure 6. Best-fit profiles of all broad lines, oversampled and shifted to a common center. The top panel shows hydrogen recombination ($H\alpha$, black, and Paschen series), the middle panel shows O I, and the bottom panel shows He I. Shaded regions indicate 1σ uncertainties for the least-constrained lines (Pa δ , O I $\lambda 11290$, and He I $\lambda 6680$); uncertainties for the remaining lines are omitted for clarity. We note that hydrogen and O I lines show largely similar widths, consistent with their coupling through charge exchange. By contrast, He I lines are systematically narrower, likely reflecting their distinct metastable triplet physics and formation in a less dense outer region.

velocity offsets between various line species in the spectrum of GLIMPSE-17775. As mentioned previously, we choose the redshift of the narrow forbidden [S III] $\lambda 9701$ as the systemic reference, due to it being relatively bright and isolated. We place all velocity offsets in context by plotting all values from Tables 4 and 5 in Figure 7.

While every fit formally yields a centroid, the significance of any offset is constrained by the spectral resolution and S/N of the data. To gauge which velocity shifts are meaningful, we adopt a conservative uncertainty of $\Delta V \sim \text{med}(R)/5$, with $R = \lambda/\Delta\lambda$. This corresponds to $\sim 60 \text{ km s}^{-1}$, below which offsets are unlikely to be statistically significant.

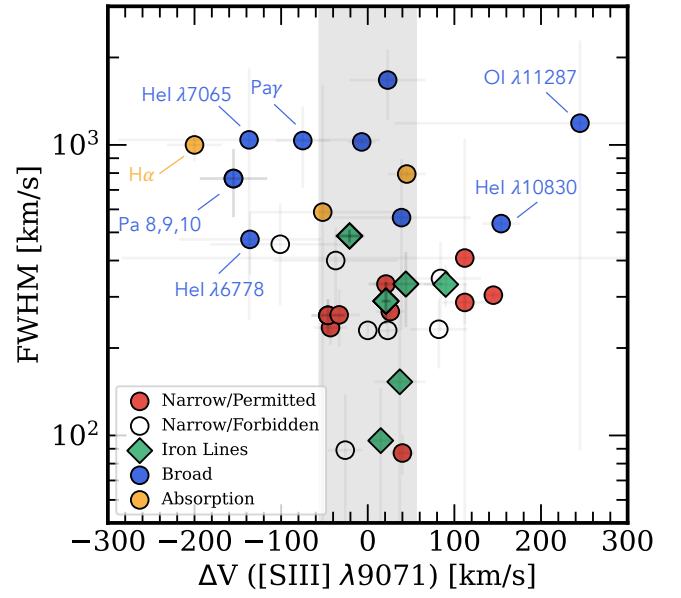


Figure 7. Velocity offsets (relative to [S III] $\lambda 9071$) and FWHM for all detected lines. Narrow permitted lines are shown as maroon circles (open for forbidden), broad lines as blue, absorption as gold, and Fe II lines as green diamonds. The shaded band marks the velocity uncertainty set by the median spectral resolution. Narrow lines align closely with the systemic redshift, while absorption features show moderate blueshifts of $\sim 150 \text{ km s}^{-1}$.

Qualitatively, we do not find systematic shifts of either permitted or forbidden narrow lines relative to the systemic redshift. Iron lines also appear consistent with the systemic velocity. In contrast, the permitted broad lines, together with their associated absorption components, show a consistent blueshift of order $\sim 100 \text{ km s}^{-1}$.

Such blueshifts are commonly interpreted as signatures of outflowing gas in the BLR, where scattering and absorption occur preferentially along outflowing sight lines. In the context of GLIMPSE-17775, this modest but systematic offset strengthens the case for a stratified cocoon, while the narrow lines trace gas in the outer regions at systemic velocity, the BLR and absorbing layers appear to be participating in bulk outflows. Confirming the detailed velocity structure, however, will require higher-resolution follow-up with the NIRSpc H-gratings (e.g., A. Saldana-Lopez et al. 2025; A. Torralba et al. 2026).

5.4. Hydrogen Absorption Lines

Although no strong hydrogen absorption lines are explicitly detected, the $H\alpha$ profile (Figure 2) shows a noticeable asymmetry, with suppressed flux on the blue side of the line. This feature is best modeled by a blueshifted ($\sim -200 \text{ km s}^{-1}$) absorption component, as already noted in Section 4. Similar nonresonant absorption signatures (e.g., Balmer lines) are now routinely observed in LRDs across a wide range of redshifts (e.g., J. E. Greene et al. 2024; V. Kokorev et al. 2024b; I. Labbe et al. 2024; X. Lin et al. 2024; A. J. Taylor et al. 2025). The presence of such absorption implies high gas densities of order $n \sim 10^9 \text{ cm}^{-3}$ (P. B. Hall 2007; K. Inayoshi & R. Maiolino 2025). Interestingly, only atoms in the $n = 2$ state appear significantly populated, as we detect no evidence for Paschen absorption in any of the five lines present in our spectrum. This is not an instrumental effect of the M-grating, as we clearly detect an absorption feature in He I $\lambda 10830$ but

not in the immediately adjacent Pa γ . This indicates that the $n = 3$ state is comparatively underpopulated.

Another key manifestation of the same physical mechanism is the Balmer break, corresponding to the $n = \infty \rightarrow n = 2$ transition limit. A prominent break is common in many LRD spectra (D. J. Setton et al. 2025), though not ubiquitous (e.g., V. Kokorev et al. 2023; R. Tripodi et al. 2025). While our spectral coverage does not extend to the Balmer limit itself, we observe a discontinuity of ~ 1 mag between the F200W and F150W filters. Due to the F200W coverage of the SED, this flux jump cannot be attributed to line boosting from H β and the [O III] doublet. Adopting the break parameterization of R. P. Naidu et al. (2025), we use our best-fit SED and measure $f_{\nu,4050\text{\AA}}/f_{\nu,3670\text{\AA}} = 2.02 \pm 0.10$. Although weaker than the extreme values (~ 4 – 7) reported for LRDz9 (A. J. Taylor et al. 2025), MoM BH[−]1 (R. P. Naidu et al. 2025), or “the Cliff” (A. de Graaff et al. 2025b), this value lies close to the maximum achievable by evolved stellar populations (I. Labbe et al. 2024; B. Wang et al. 2024). However, this measurement is based on the total (PS+host) photometry. Given that the UV continuum is spatially resolved, as we have shown in Section 4.1, host-galaxy contamination may dilute the intrinsic break strength of the central component. While uncertainties on the PS/host decomposition increase significantly toward the rest-optical, we find that using the PS-only photometry produces a noticeably sharper break, increasing the inferred strength by $\sim 30\%$ – 50% to $f_{\nu,4050\text{\AA}}/f_{\nu,3670\text{\AA}} \approx 2.6$ – 3.0 . Although this does not allow us to robustly conclude whether the intrinsic PS break reaches the most extreme BH[−]-type objects (A. de Graaff et al. 2025b; R. P. Naidu et al. 2025; A. J. Taylor et al. 2025), it suggests that the Balmer discontinuity becomes stronger once the host contribution is removed, consistent with recent findings that host light can suppress the apparent break amplitude (e.g., A. de Graaff et al. 2025d; W. Q. Sun et al. 2026).

Taken together, the presence of asymmetry in the H α profile, likely caused by blueshifted Balmer absorption, and a Balmer break adds to the evidence, complementing the exponential line wings, that the emission in GLIMPSE-17775 arises from a dense, ionized cocoon of gas. The lack of a detectable Paschen break is consistent with this picture, as high densities and optical depths result in Thomson scattering while also naturally washing out higher-order continuum edges (e.g., B. Wang et al. 2025). Assuming LTE, an extremely low ratio $n_3/n_2 \lesssim 0.01$ corresponds to an electron temperature of $T_e \sim 5000$ K or below, consistent with the warm, partially ionized gas expected in dense LRD cocoons.

We also observe moderately prominent and extremely prominent blueshifted absorption, respectively, in He I $\lambda 7065$ and He I $\lambda 10830$. The mechanism for this is similar, but not identical to, the hydrogen lines. We also note the prominent blueshifted absorption seen in both He I $\lambda 7065$ and He I $\lambda 10830$. However, the physical mechanism driving these helium features differs fundamentally from that of the hydrogen lines; we return to this in the following section.

5.5. Helium Lines

Both He I $\lambda 7065$ and He I $\lambda 10830$ show blueshifted (~ 70 – 80 km s^{−1} from the narrow-line center) absorption components with widths of ~ 600 – 700 km s^{−1}. Unlike the Balmer series, helium transitions are resonantly scattered, so the arguments regarding $n = 2$ state abundances do not apply.

Before addressing the absorption itself, it is useful to consider what the observed line strengths already tell us.

In the spectrum of GLIMPSE-17775, we detect three He I lines: $\lambda 6680$, $\lambda 7065$, and $\lambda 10830$. These originate from different physical mechanisms. The $\lambda 6680$ singlet line is produced via recombination or radiative pumping, and is only weakly dependent on density. By contrast, $\lambda 7065$ and $\lambda 10830$ belong to the triplet system (along with $\lambda 3889$, not covered by G395M), where the lower level is populated due to its very long lifetime. Electrons in this “ground state” can be collisionally or radiatively excited into higher energies, and therefore the triplet-line strengths depend strongly on density and, to a lesser extent, temperature. As emphasized by D. A. Berg et al. (2026), He I $\lambda 10830$ is the most sensitive density diagnostic among these transitions, followed by $\lambda 7065$. The fact that both are much stronger than $\lambda 6680$ indicates that the region where they form is very dense. The prominence of $\lambda 10830$ in particular already points to a high-density environment with a temperature $T \gtrsim 10^4$ K. Further, the broad $\lambda 6680$ line has a noticeably larger FWHM than the triplet lines but is consistent with hydrogen recombination lines, which again makes sense since both originate from the same mechanism.

What about absorption? Its presence further reinforces this picture. The lower level (2^3S) of the triplet system is a metastable, long-lived state that effectively acts as a ground state in dense or partially ionized gas. As a result, photons from the $2^3P \rightarrow 2^3S$ transition can be resonantly absorbed and reemitted many times, analogous to Ly α or Mg II. This mechanism naturally produces strong absorption features when the He I column density is high. The fact that He I $\lambda 10830$ not only dominates the helium spectrum in emission but also shows the deepest absorption in the entire spectrum provides compelling evidence for a dense, ionized cocoon of gas enshrouding GLIMPSE-17775.

5.6. Oxygen Lines

Above, we noted that O I $\lambda 8446$ and $\lambda 11290$ share very similar profiles with the hydrogen recombination lines. This is naturally expected because neutral oxygen is tightly coupled to hydrogen via charge exchange, which rapidly equilibrates the O/H ionization states in dense gas. When charge exchange is fast (high n_H), the O I/O II ratio tracks the H I/H II ratio, so the O I-emitting gas shares the same kinematics as the hydrogen recombination region (B. T. Draine 2011). The observed agreement in FWHM between H and O I lines is therefore an important consistency check that both species originate in the same dense, partially ionized phase.

A second, independent clue of high gas density and radiation field intensity comes from the excitation mechanism. The O I $\lambda 8446$ – $\lambda 11287$ pair is the classic signature of Ly β (Bowen) fluorescence: Ly β pumps O I from the ground state and the ensuing cascade preferentially populates the levels that emit at $1.129 \mu\text{m}$, which then cascade down through the $8446 \text{ \AA}$ line. In this channel, the line emissivity scales with both the neutral oxygen (or equivalently neutral hydrogen, via charge exchange) column and the local Ly β radiation field. Therefore, producing strong O I fluorescence requires both a very bright Ly β source and a dense neutral (or partially ionized) cocoon.

Further, we do not detect (5σ upper limit $< 12/[10^{-20} \text{ erg s}^{-1} \text{ cm}^{-2}]$) any of the other permitted

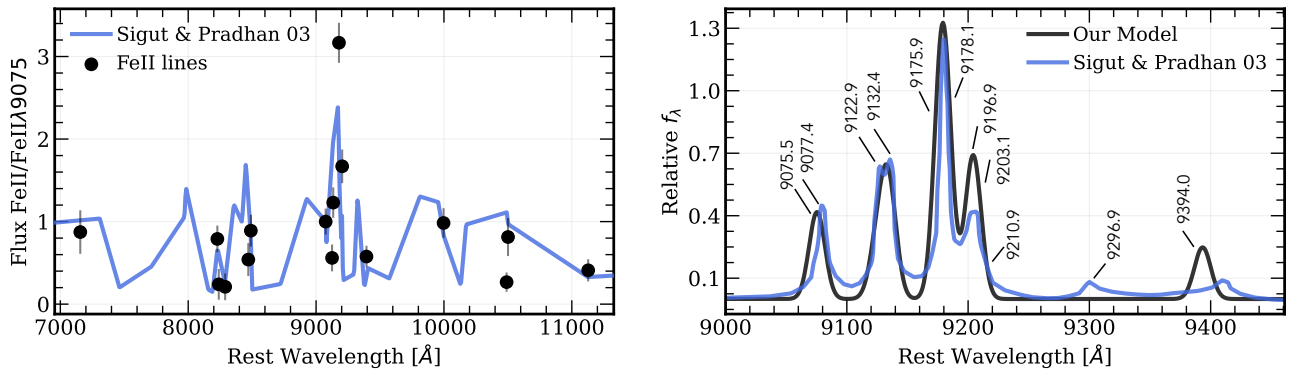


Figure 8. Ly α -pumped iron emission in GLIMPSE-17775. Left: observed Fe II flux ratios (black points), normalized to Fe II λ 9075, compared with predictions from Ly α fluorescence models of T. A. A. Sigut & A. K. Pradhan (2003, blue). The close correspondence across ~ 14 lines indicates a common excitation mechanism driven by Ly α pumping in dense, partially shielded gas. Right: zoom in on the $\lambda\lambda$ 9000–9400 complex, showing the remarkable agreement between the modeled continuum-subtracted Fe II spectrum (black) and the T. A. A. Sigut & A. K. Pradhan (2003) model (blue). We show air wavelengths of each Fe II line directly from T. A. A. Sigut & A. K. Pradhan (2003), including the blended features. Together, these comparisons support an origin in the dense inner regions of the cocoon surrounding the accreting black hole.

O I λ 7774, λ 7254, and λ 7790 lines, and the [O I] λ 6302/O I λ 8446 ratio is weak. If collisional excitation or pure recombination dominated, these lines would be comparatively strong; their absence strongly favors Bowen fluorescence as the primary driver of the observed O I emission.

Finally, the velocity structure adds a natural stratification: hydrogen recombination and O I (fluorescent, charge-exchange coupled) share similar widths and profiles, while He I lines are systematically narrower. Since He I triplet transitions emerge from a metastable 2^3S level (with distinct, density-sensitive population pathways and resonant transfer), they likely trace a kinematically distinct layer within the same cocoon.

5.7. Ly α Fluorescence in Iron Lines

Permitted iron emission, primarily Fe II and Fe III (e.g., I. Labbe et al. 2024; R. Tripodi et al. 2025; A. Torralba et al. 2026), but in some cases extending to highly ionized species such as Fe VII (E. Lambrides et al. 2025; M. Tang et al. 2025) and Fe X (L. J. Furtak et al. 2024), has become a recurring feature in LRD spectra. In GLIMPSE-17775, the spectrum is exceptionally rich in NIR iron lines; we identify 16 Fe II and one Fe III transition. Understanding the origin of this emission is key to interpreting the dense-gas environment in LRDs.

The physics of Fe II emission has long been a challenge for BLR photoionization models (M. Joly 1993). Thick, high-column-density ($\sim 10^{25} \text{ cm}^{-2}$) gas at the edges of the accretion disk has been invoked as a potential source (M. Joly 1987; S. Collin-Souffrin et al. 1988), where the scattering and absorption of the hard X-ray photons ionize the gas. Similarly, this would also enhance the Balmer and Paschen line luminosities. Further, extensive theoretical work has shown that Ly α fluorescence is fundamental in reproducing the observed Fe II strengths (T. A. A. Sigut & A. K. Pradhan 1998, 2003). Ly α pumping not only boosts the UV and optical Fe II emission, but also predicts strong lines in the NIR.

This expectation aligns closely with our observations. In Figure 8, we compare our measured Fe II flux ratios (normalized to Fe II λ 9075) against the Ly α -pumped model predictions of T. A. A. Sigut & A. K. Pradhan (2003). With the exception of Fe II λ 9179, affected by blending with Fe II λ 9204 and Pa9, we find remarkably good agreement across the suite of detected features. In particular, the dense

forest of Fe II transitions spanning $\lambda\lambda$ 9000–9200 (center-right panel in Figure 2) is reproduced almost exactly by the theoretical spectrum (see Figure 11 in T. A. A. Sigut & A. K. Pradhan 2003). We overlay the T. A. A. Sigut & A. K. Pradhan (2003) model directly on our continuum-subtracted fit in Figure 8, demonstrating that the observed Fe II emission is fully consistent with fluorescence in dense gas near the BLR. The implication is that the majority of the Fe II emission in GLIMPSE-17775 arises through Ly α pumping, directly tracing an extremely dense medium, ionized by an extremely luminous source, likely an accreting black hole. Given that the Fe II lines are somewhat narrower than other permitted features, it is likely that the emitting region is located farther from the BLR—something that has already been shown in a classic example of a narrow Fe II emitter, I Zw 1 (R. J. Rudy et al. 2000; A. O. M. Marinello et al. 2016).

Together with the Balmer absorption, helium triplet features, and Bowen fluorescence in O I, the iron forest adds another piece of evidence that points to a dense cocoon enshrouding the black hole.

5.8. Source Properties

Based on the previous discussion, we now present the likely physical properties of the black hole and its host. The coexistence of broad permitted and narrow forbidden lines indicates, as in most LRDs (R. E. Hviding et al. 2025), that the broad components originate in an AGN BLR. Following J. E. Greene & L. C. Ho (2005), we derive M_{BH} from the luminosity and width of the broad H α line, assuming the exponential-wing profile demonstrated in Figures 2 and 5. With a $\text{FWHM}_{\text{H}\alpha} = 1024 \pm 21 \text{ km s}^{-1}$, negligible dust, and $\mu = 2.04$, we obtain $\log(M_{\text{BH}}/M_{\odot}) = 6.65 \pm 0.15$, where the dominant uncertainty arises from the J. E. Greene & L. C. Ho (2005) calibration. A purely Gaussian fit, though statistically disfavored, would yield a FWHM 3 times larger and a black hole mass nearly a dex higher. Throughout this work, we adopt the M_{BH} from the exponential model.

Further, by assuming that the bolometric luminosity (L_{bol}) scales as $L_{\text{bol}} = 130 \times L_{\text{H}\alpha}$ (G. T. Richards et al. 2006), we find $L_{\text{bol}} = (1.06 \pm 0.14) \times 10^{45} \text{ erg s}^{-1}$. Using our M_{BH} derived by assuming exponential wings, we find that this object is accreting at a super-Eddington rate, $L_{\text{bol}}/L_{\text{edd}} = 1.86 \pm 0.25$. This is higher than the vast majority of LRDs at all redshifts

(e.g., V. Kokorev et al. 2023; L. J. Furtak et al. 2024; I. Juodžbalis et al. 2024; R. Maiolino et al. 2024; H. B. Akins et al. 2025b; D. D. Kocevski et al. 2025; A. J. Taylor et al. 2025), found to accrete at sub-Eddington rates. However, recent works examining LRDs with extreme Balmer breaks derive accretion rates that exceed the Eddington limit (E. Lambrides et al. 2024; A. de Graaff et al. 2025b; R. P. Naidu et al. 2025).

A recent work examining multiwavelength LRD data suggests that the bolometric luminosity emerges predominantly from the rest-frame optical, with X-ray and radio contributions being largely subdominant (J. E. Greene et al. 2026). Given the suggested dominance of optical light in LRDs, the much lower bolometric corrections ($\times 7$ – 10 lower) would imply lower black hole masses and total bolometric luminosities (e.g., those derived from $H\alpha$) than standard AGN prescriptions (e.g., J. E. Greene & L. C. Ho 2005) might suggest. This adjustment would lower both the inferred black hole masses and bolometric luminosities. Because these two quantities scale together, the implied super-Eddington nature of GLIMPSE-17775 would remain unchanged, although the black hole mass could decrease to $\sim 10^{5.5}$ – $10^{5.8} M_\odot$. This interpretation is consistent with dense-gas (BH*) models, which generically predict Eddington or super-Eddington accretion in heavily obscured environments (e.g., D. Kido et al. 2025; H. Liu et al. 2025; A. Sneppen et al. 2026). A more detailed reassessment of bolometric corrections for LRDs is clearly warranted, but such an analysis lies beyond the scope of this work. For consistency with previous studies, we adopt standard AGN bolometric corrections throughout, with all derived parameters listed in Table 1.

Finally, we estimate an upper limit on the stellar mass in GLIMPSE-17775. Previous studies have modeled LRDs using joint galaxy+AGN SED decomposition (e.g., V. Kokorev et al. 2023; L. J. Furtak et al. 2024), and more recently, within the BH* framework, have assumed that most of the rest-UV flux arises from the host galaxy (R. P. Naidu et al. 2025; A. J. Taylor et al. 2025). Our spatial decomposition (Figure 4) supports this assumption: The extended component contributes $>80\%$ of the rest-UV light, implying that the bulk of the stellar mass resides in this resolved structure. Given this, we adopt a simple empirical approach using the M_{UV} – M_* relation (e.g., D. P. Stark et al. 2009; I. Labbé et al. 2013) and obtain a conservative upper limit on the stellar mass of $M_* \lesssim 10^{7.5} M_\odot$. This in turn gives us a black hole-to-host mass ratio of $\lesssim 0.14$, which is significantly elevated from local expectations (J. E. Greene & L. C. Ho 2005), but is not as extreme as some other LRDs reported in the literature (V. Kokorev et al. 2023; L. J. Furtak et al. 2024; R. Maiolino et al. 2024).

6. Discussion and Summary

6.1. Dense and Ionized Gas Surrounding the AGN

Our NIRSpect G395M observations of GLIMPSE-17775 reveal a remarkably consistent picture: Across independent tracers, the line emission requires an environment of extremely high density and partial ionization.

First, the broad permitted lines are universally better fit by exponential wings, a hallmark of electron scattering in gas with $n_e \gtrsim 10^8 \text{ cm}^{-3}$ and column densities approaching $N_e \sim 10^{24} \text{ cm}^{-2}$. Such profiles are not reproduced by Doppler broadening alone and point to an ionized scattering medium enveloping the source. The fact that hydrogen and O I lines

share consistent FWHM values further anchors this interpretation, as charge exchange tightly couples neutral oxygen to the ionization and kinematics of hydrogen.

Second, the detection of blueshifted Balmer absorption and a significant Balmer break both require high $n = 2$ populations and densities $n \sim 10^9 \text{ cm}^{-3}$. Helium transitions provide a complementary view: The triplet lines $\lambda 7065$ and $\lambda 10830$ are both strongly enhanced relative to the singlet states and show deep blueshifted absorption, consistent with resonant scattering from the metastable 2^3S level. Their systematically narrower widths compared to hydrogen and oxygen suggest stratification, with helium arising from denser, more compact layers of the cocoon.

Third, the O I $\lambda 8446$ – $\lambda 11290$ pair confirms Ly β fluorescence, requiring both a bright Ly β radiation field and a dense reservoir of neutral gas. The absence of other permitted O I lines not fed by Ly β strengthens this conclusion. Finally, the detection of 16 Fe II lines, forming an incredibly rich iron forest in this LRD, matches predictions from Ly α fluorescence models, again demanding an intense radiation field and very high densities.

Because the BLR is unresolved in essentially all AGN, broad-line widths are traditionally interpreted as virial tracers of the black hole potential. In GLIMPSE-17775, the virial story alone is insufficient: The issue is not how broad the lines are, but how they broaden. The profiles exhibit extended, nearly linear wings in velocity space that are incompatible with a Gaussian. Instead, the lines are systematically and significantly better described by a model consisting of a narrow Gaussian core (virial motion) plus exponential wings, the hallmark of Thomson scattering in a dense ionized medium. Thus, the line shape encodes both gravitational kinematics and radiative-transfer physics in the surrounding cocoon. Our schematic (Figure 9) summarizes this revised view: In LRDs, broad-line profiles are not set by dynamics alone, but by the scattering environment through which the photons escape.

Such conditions—high optical depths, large column densities, and evidence for radiation-dominated gas—are precisely those expected in super-Eddington accretion flows. In this regime, radiation pressure inflates the inner accretion structure, driving powerful winds and forming the very dense, partially ionized envelope we infer here. The low X-ray luminosities and weak radio emission commonly observed in LRDs (e.g., T. T. Ananna et al. 2024; M. Yue et al. 2024; H. B. Akins et al. 2025a; M. Kokubo & Y. Harikane 2025) are consistent with this picture: The X-rays are likely absorbed or thermalized within the optically thick cocoon, while dust cannot survive in such an intense radiation field (e.g., E. Lambrides et al. 2024; A. de Graaff et al. 2025b; K. Inayoshi & R. Maiolino 2025; R. P. Naidu et al. 2025; A. J. Taylor et al. 2025). Thus, the spectroscopic signatures observed in GLIMPSE-17775—exponential wings, absorption features, and fluorescence-driven metal lines—fit naturally into a scenario where super-Eddington accretion onto a low-mass black hole powers the luminous yet heavily reprocessed emission.

Taken together, the exponential wings, Balmer and helium absorption, Bowen oxygen lines, and Ly α -pumped iron forest all converge on the same physical picture: GLIMPSE-17775 is enshrouded in a dense, partially ionized cocoon of gas.

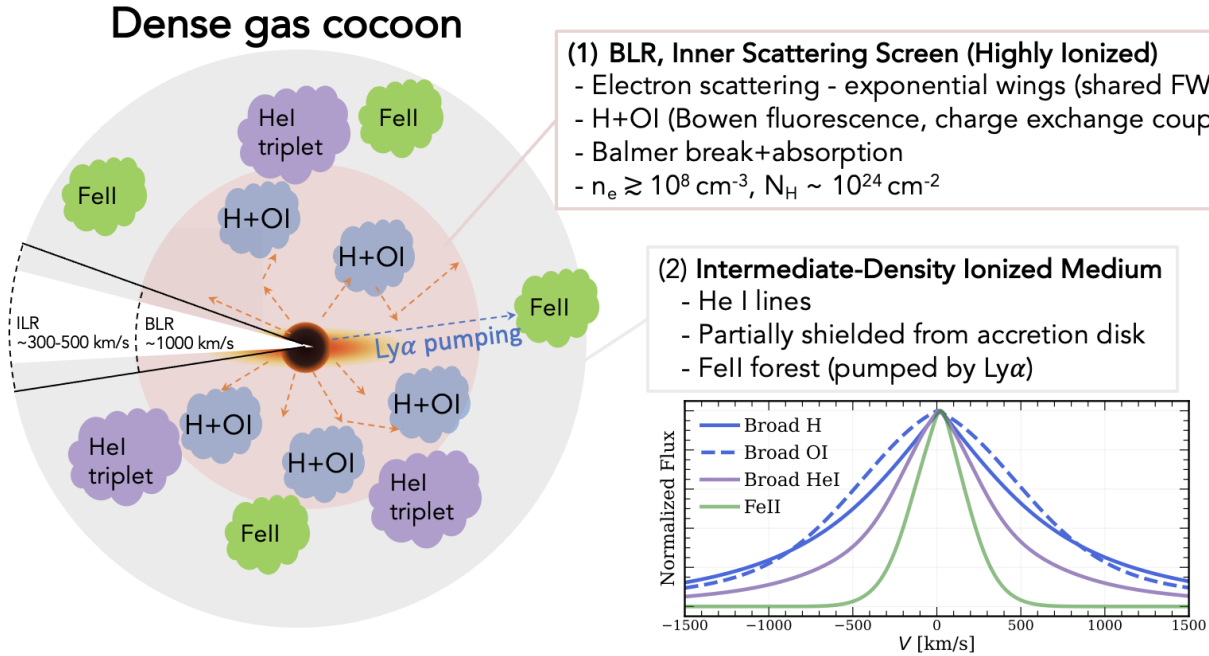


Figure 9. Physical picture of the dense cocoon around GLIMPSE-1775. The inner region (1) corresponds to the dense, highly ionized broad-line region (BLR) where electron scattering (orange lines) produces exponential wings, H and O I share coupled kinematics (via charge exchange and Bowen fluorescence), and strong Balmer absorption/break signatures arise. The outer layer (2) represents an intermediate-density ionized medium, where He I triplet lines show resonant absorption from the metastable 2^3S level and Fe II emission is driven by $\text{Ly}\alpha$ fluorescence. Together, these zones form a stratified cocoon enshrouding the accreting black hole. The bottom-right panel shows stacked line profiles of H, O I, He I, and Fe II.

6.2. Final Remarks

Using a combination of the intrinsically deepest NIRCам photometry and NIRSpec G395M observations of the lensed AS1063 field, we present a detailed investigation of a LRD at $z = 3.501$. The spectrum of GLIMPSE-1775 is exceptionally rich, with more than 40 detected features, allowing us to probe the physical conditions of the gas with unprecedented detail. Multiple independent diagnostics converge on the presence of a dense, partially ionized cocoon heated by a powerful ionizing source.

Typical of other LRDs, GLIMPSE-1775 is extremely compact in the rest-optical ($r_{\text{eff}} < 300 \text{ pc}$) but exhibits more extended structure in the rest-UV ($r_{\text{eff}} \sim 1000 \text{ pc}$; Figure 1). It shows unmistakable AGN signatures through broad permitted lines (e.g., D. D. Kocevski et al. 2023; V. Kokorev et al. 2023; J. E. Greene et al. 2024; L. J. Furtak et al. 2024). The uniquely deep G395M spectrum further reveals clues to the physical origin of both its continuum and line-emission properties: We detect exponential broad-line wings, Balmer and helium absorption, Bowen-pumped O I, Fe II emission produced by $\text{Ly}\alpha$ fluorescence, and evidence for rapid, potentially super-Eddington growth.

Together, these diagnostics provide one of the clearest cases yet for the BH* “dense cocoon” scenario (A. de Graaff et al. 2025b; K. Inayoshi & R. Maiolino 2025; R. P. Naidu et al. 2025; A. J. Taylor et al. 2025). The same ingredients we observe—exponential broad wings, Balmer-break absorption, He I absorption, and rich Fe II and O I fluorescence—have been detected individually in several other bright LRDs with medium- and high-resolution spectroscopy (e.g., L. J. Furtak et al. 2024; I. Juodžbalis et al. 2024; I. Labbe et al. 2024; F. D’Eugenio et al. 2025; E. Lambrides et al. 2025; B. Wang et al. 2025; A. Torralba et al. 2026), suggesting that dense,

optically thick gas may be a common feature of the population rather than an anomaly. What distinguishes GLIMPSE-1775 is that all signatures are captured simultaneously and at high S/N, allowing a self-consistent physical interpretation. If such cocoons are widespread, then super-Eddington accretion may be a typical pathway for black hole growth in LRDs, especially among the most luminous systems. Establishing how these signatures vary with luminosity and redshift will be essential for determining whether dense cocoons represent a dominant mode of early black hole assembly.

The convergence of five independent diagnostics—exponential scattering wings, Balmer-limit absorption, helium triplet physics, and two distinct fluorescence channels—leaves little doubt: GLIMPSE-1775 hosts a dense ($n \sim 10^{8-9} \text{ cm}^{-3}$), optically thick ($N_e \sim 10^{24} \text{ cm}^{-2}$) cocoon of partially ionized gas surrounding a super-Eddington accreting black hole. This represents some of the most direct and comprehensive spectroscopic evidence to date for the dense cocoon scenario in LRDs.

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Facilities: JWST, HST.

Software: EZY (G. B. Brammer et al. 2008), grizli (G. Brammer 2023), msaexp (G. Brammer 2022), photutils (L. Bradley et al. 2020), pysersic (I. Pasha & T. B. Miller 2023), sep (K. Barbary 2016), SExtractor (E. Bertin & S. Arnouts 1996).

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